

TRIGONOMETRY

(LECTURE – 2)

COMPARING COEFFICIENTS

If $(a^2+b^2)\sin x + 2ab \cos x = a^2+b^2$, then find $\tan x$?

A) $(a^2-b^2)/2ab$

B) $2ab/(a^2-b^2)$

C) $(a^2-b^2)/(a^2+b^2)$

D) $(a^2+b^2)/(a^2-b^2)$

**$2\sqrt{a} \cos x + (1-a) \sin x = (1+a)$,
find $\cot x$?**

- A) $(1-a)/2\sqrt{a}$**
- B) $(1+a)/2\sqrt{a}$**
- C) $2\sqrt{a}/(1+a)$**
- D) $2\sqrt{a}/(1-a)$**

If $24 \sin^4 x + 40 \operatorname{cosec}^4 x = 15$,

Find $54 \tan^6 x + 125 \operatorname{cosec}^6 x$?

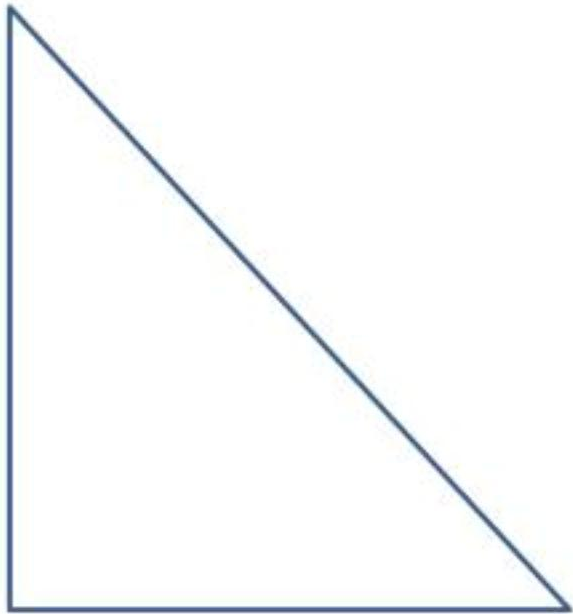
A) 729

B) 762

C) 716

D) 625

Important triplet used in Trigo :-



Concept :-

$$a^2 - b^2 = (a+b)(a-b)$$

$$\sec^2 x - \tan^2 x = 1$$

Therefore,

$$\sec x + \tan x = 1/(\sec x - \tan x)$$

$$\operatorname{cosec}^2 x - \cot^2 x = 1$$

Therefore

$$\operatorname{cosec} x + \cot x = 1/(\operatorname{cosec} x - \cot x)$$

$$\sec^2 x - \tan^2 x = 1$$

Therefore,

$$\sec x + \tan x = 1/(\sec x - \tan x)$$

Example : 1

If $\sec \theta - \tan \theta = \frac{x}{y}$, ($0 < x < y$) and ($0^\circ < \theta < 90^\circ$), then $\sin \theta = ?$

यदि $\sec \theta - \tan \theta = \frac{x}{y}$, ($0 < x < y$) व ($0^\circ < \theta < 90^\circ$) हो, तो $\sin \theta = ?$

SSC CGL 4th March 2020 (morning)

A. $\frac{x^2+y^2}{2xy}$

B. $\frac{2xy}{x^2+y^2}$

C. $\frac{y^2-x^2}{x^2+y^2}$

D. $\frac{y^2+x^2}{y^2-x^2}$

Example : 2

If $\sec \theta + \tan \theta = p$ ($0^\circ < \theta < 90^\circ$), then $\frac{p^2-1}{p^2+1} = ?$

यदि $\sec \theta + \tan \theta = p$ ($0^\circ < \theta < 90^\circ$), हो, तो $\frac{p^2-1}{p^2+1} = ?$

SSC CGL March 4 2020 (evening)

A. $\operatorname{cosec} \theta$

B. $\sin \theta$

C. $\cos \theta$

D. $2\operatorname{cosec} \theta$

$$\operatorname{Cosec}^2 x - \cot^2 x = 1$$

Therefore

$$\operatorname{Cosec} x + \cot x = 1/(\operatorname{cosec} x - \cot x)$$

Example

If $x = \operatorname{cosec} A + \cot A$, $y = \operatorname{cosec} A - \cot A$,

Find $\frac{1}{1+x^2} + \frac{1}{1+y^2}$

CROSS MULTIPLICATION METHOD USED IN TRIGO:-

$$a_1x + b_1y + c_1 = 0$$

$$a_2x + b_2y + c_2 = 0$$

$$x = \frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1},$$

$$y = \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1}$$

X		Y		1	
Coefficient of Y	Constant term	Constant term	Coefficient of X	Coefficient of X	Coefficient of Y
$\begin{array}{cc} b_1 & \nearrow c_1 \\ b_2 & \searrow c_2 \end{array}$		$\begin{array}{cc} c_1 & \nearrow a_1 \\ c_2 & \searrow a_2 \end{array}$		$\begin{array}{cc} a_1 & \nearrow b_1 \\ a_2 & \searrow b_2 \end{array}$	

$$\frac{X}{b_1c_2 - b_2c_1} = \frac{Y}{c_1a_2 - c_2a_1} = \frac{1}{a_1b_2 - a_2b_1}$$

12. If $a \sec \theta + b \tan \theta + c = 0$,

$$p \sec \theta + q \tan \theta + r = 0,$$

find $(br - qc)^2 - (pc - ar)^2$.

- a) $(qa + bp)^3$
- b) $(aq - bp)^3$
- c) $(aq + bp)^2$
- d) $(aq - bp)^2$

$$\text{if } a\operatorname{cosec} + b\cot + 5 = 0$$

$$x\operatorname{cosec} + y\cot + 2 = 0$$

$$\text{Find } (2b-5y)^2 - (5x-2a)^2$$

If $\operatorname{cosec} \theta = 3x$ and $\cot \theta = \frac{3}{x}$, ($x \neq 0$), then $6\left(x^2 - \frac{1}{x^2}\right) = ?$

यदि $\operatorname{cosec} \theta = 3x$ व $\cot \theta = \frac{3}{x}$, ($x \neq 0$) हो, तो $6\left(x^2 - \frac{1}{x^2}\right) = ?$

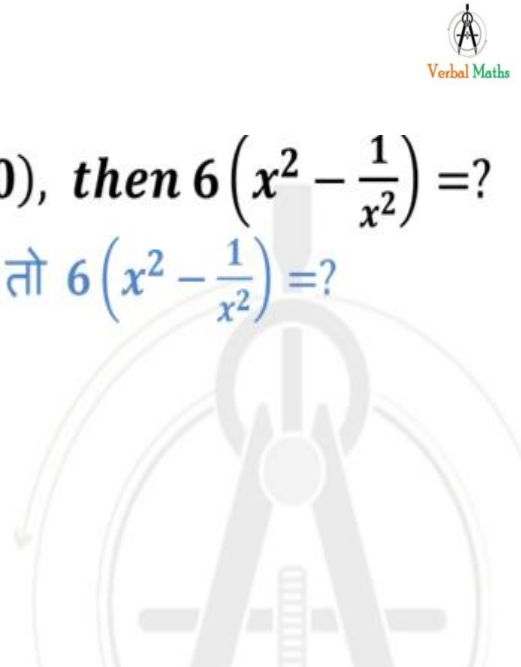
SSC CHSL July 8 2019 (morning)

A. $2/3$

B. 1

C. $1/4$

D. $1/2$



If $\sin \theta = 3x$ and $\cos \theta = \frac{3}{x}$, ($x \neq 0$), then $\left(x^2 + \frac{1}{x^2}\right) = ?$

यदि $\sin \theta = 3x$ व $\cos \theta = \frac{3}{x}$, ($x \neq 0$) हो, तो $6 \left(x^2 + \frac{1}{x^2}\right) = ?$

SSC CHSL July 8 2019 (afternoon)

If $\tan \theta + \sec \theta = \frac{(x-2)}{(x+2)}$, then $\cos \theta = ?$

यदि $\tan \theta + \sec \theta = \frac{(x-2)}{(x+2)}$ हो, तो $\cos \theta = ?$

A. $\frac{x^2-1}{x^2+1}$

B. $\frac{(2x^2+4)}{(2x^2-4)}$

C. $\frac{x^2-4}{x^2+4}$

D. $\frac{x^2-2}{x^2+2}$



If $\sec \theta = \frac{a}{b}, b \neq 0$, then $\frac{1-\tan^2 \theta}{2-\sin^2 \theta} = ?$

यदि $\sec \theta = \frac{a}{b}, b \neq 0$ हो, तो $\frac{1-\tan^2 \theta}{2-\sin^2 \theta} = ?$

A. $\frac{a^2(2b^2+a^2)}{b^2(a^2-b^2)}$

B. $\frac{a^2(2b^2+a^2)}{b^2(a^2+b^2)}$

C. $\frac{a^2(2b^2-a^2)}{b^2(a^2+b^2)}$

D. $\frac{b^2(2b^2-a^2)}{a^2(a^2+b^2)}$





If $\operatorname{cosec} \theta = \frac{b}{a}$, then $\frac{\sqrt{3} \cot \theta + 1}{\tan \theta + \sqrt{3}} = ?$

यदि $\operatorname{cosec} \theta = \frac{b}{a}$ हो, तो $\frac{\sqrt{3} \cot \theta + 1}{\tan \theta + \sqrt{3}} = ?$

A. $\frac{\sqrt{b^2 - a^2}}{a}$

B. $\frac{\sqrt{b^2 + a^2}}{a}$

C. $\frac{\sqrt{b^2 + a^2}}{b}$

D. $\frac{\sqrt{b^2 - a^2}}{b}$



If $\cot \theta = \frac{\sin \alpha - \cos \alpha}{\sin \alpha + \cos \alpha}$, then find $(\sin \alpha + \cos \alpha)$.

a. $\pm \sqrt{2} \cos \theta$

b. $\pm \sqrt{2} \sin \theta$

c. $\pm \sqrt{2} \sec \theta$

d. $\pm \sqrt{2} \operatorname{cosec} \theta$



If $\tan\theta = \frac{\sin\alpha - \cos\alpha}{\sin\alpha + \cos\alpha}$, then find $(\sin\alpha + \cos\alpha)$.

a. $\pm\sqrt{2}\cos\theta$

b. $\pm\sqrt{2}\sin\theta$

c. $\pm\sqrt{2}\sec\theta$

d. $\pm\sqrt{2}\operatorname{cosec}\theta$

If $\frac{(\sin \theta - \operatorname{cosec} \theta)(\cos \theta - \sec \theta)}{\tan^2 \theta - \sin^2 \theta} = r^3$, then $r = ?$

यदि $\frac{(\sin \theta - \operatorname{cosec} \theta)(\cos \theta - \sec \theta)}{\tan^2 \theta - \sin^2 \theta} = r^3$ हो, तो $r = ?$

- A. $\sin \theta \cos \theta$
- B. $\tan \theta$
- C. $\cot \theta$
- D. $\operatorname{cosec} \theta \sec \theta$



If $\sqrt{\frac{1-\cos \theta}{1+\cos \theta}} \times \sqrt{\frac{\operatorname{cosec} \theta - \cot \theta}{\operatorname{cosec} \theta + \cot \theta}} = \frac{1-r}{1+r}$, then $r = ?$

यदि $\sqrt{\frac{1-\cos \theta}{1+\cos \theta}} \times \sqrt{\frac{\operatorname{cosec} \theta - \cot \theta}{\operatorname{cosec} \theta + \cot \theta}} = \frac{1-r}{1+r}$ हो, तो $r = ?$

- A. $\sin \theta$
- B. $\operatorname{cosec} \theta$
- C. $\sec \theta$
- D. $\cos \theta$



$$\frac{(1+\cos \theta)^2+\sin^2 \theta}{(\operatorname{cosec}^2 \theta-1) \sin^2 \theta}=?$$

- A. $\cos \theta(1+\sin \theta)$
- B. $2 \cos \theta(1+\sec \theta)$
- C. $\sec \theta(1+\sin \theta)$
- D. $2 \sec \theta(1+\sec \theta)$

Thankyou

TRIGONOMETRY

(LECTURE – 3)

IMPORTANT QUESTION :

$$\frac{\sin A - \cos A + 1}{\sin A + \cos A - 1} = ?$$

OUTPUTS :

$$= 1/(\sec A - \tan A)$$

$$= (\sec A + \tan A)$$

$$= \frac{1 + \sin A}{\cos A}$$

$$= \frac{\sin \frac{A}{2} + \cos \frac{A}{2}}{\sin \frac{A}{2} - \cos \frac{A}{2}}$$

IMPORTANT QUESTION :

$$\frac{\cos A + \sin A - 1}{\cos A - \sin A + 1} = ?$$

OUTPUTS :

A. $\operatorname{cosec} A - \cot A$

B. $\frac{1 - \cos A}{\sin A}$

C. $\tan \frac{A}{2}$

D. All of these

IMPORTANT QUESTION :

$$\frac{\cos A - \sin A + 1}{\cos A + \sin A - 1} = ?$$

OUTPUTS :

A. $\operatorname{cosec} A + \cot A$

B. $\frac{1 + \cos A}{\sin A}$

C. $\cot \frac{A}{2}$

D. All of these

$$\frac{\sin A - \cos A + 1}{\sin A + \cos A - 1} = ?$$

- a) $\frac{1 + \sin A}{\cos A}$
- b) $\frac{\cos A}{1 + \sin A}$
- c) $1 + \sin A$
- d) $1 - \sin A$

$$\frac{\sin 2A - \cos 2A + 1}{\sin 2A + \cos 2A - 1} = ?$$

A. $\frac{\sin A + \cos A}{\sin A - \cos A}$

B. $\frac{\sin A + \cos A}{\cos A - \sin A}$

C. $\sin \frac{A}{2}$

D. $\cos \frac{A}{2}$

$$\frac{\cos A + \sin A - 1}{\cos A - \sin A + 1} = ?$$

A. $\operatorname{cosec} A - \cot A$

B. $\frac{1 - \cos A}{\sin A}$

C. $\tan \frac{A}{2}$

D. All of these

$$\frac{\cos A - \sin A + 1}{\cos A + \sin A - 1} = ?$$

A. $\operatorname{cosec} A + \cot A$

B. $\frac{1 + \cos A}{\sin A}$

C. $\cot \frac{A}{2}$

D. All of these

$$\frac{1 - \tan(90 - \alpha) + \sec(90 - \alpha)}{\tan(90 - \alpha) + \sec(90 - \alpha) + 1} =$$

- a. $\tan\left(\frac{\alpha}{2}\right)$
- b. $\cot\left(\frac{\alpha}{2}\right)$
- c. $\sin\left(\frac{\alpha}{2}\right)$
- d. $\cos\left(\frac{\alpha}{2}\right)$

If $\frac{(1 + \sin \theta - \cos \theta)}{(1 + \sin \theta + \cos \theta)} + \frac{(1 + \sin \theta + \cos \theta)}{(1 + \sin \theta - \cos \theta)} = 4$, then which of the following values will be suitable for θ ?

यदि $\frac{(1 + \sin \theta - \cos \theta)}{(1 + \sin \theta + \cos \theta)} + \frac{(1 + \sin \theta + \cos \theta)}{(1 + \sin \theta - \cos \theta)} = 4$ हो, तो इनमे से कोन सा मान θ के लिए उपयुक्त होगा?

- A. 30°
- B. 45°
- C. 60°
- D. 90°

If

$$\sin x = \cos y ,$$

$$\sec x = \operatorname{cosec} y,$$

$$\tan x = \cot y$$

$$\text{Then } x+y = 90^\circ$$

If $\sec 4\theta = \operatorname{Cosec} (\theta + 20^\circ)$, then θ is equal to:

If $\tan \theta = \cot (\theta + 45^\circ)$, then θ is equal to:

If $\sin 7\theta = \cos (13\theta - 45^\circ)$, then θ is equal to:

Find $\sin(75+X) - \cos(15-X) + \cot(57+3X) - \tan(33-3X) + \operatorname{cosec}(43-5X) - \sec(47+5X)$?

- 0
- 1
- 1
- 2

IMPORTANT PROTOTYPE QUESTIONS :

If $\sin^2 x + \sin x = 1$

Then $\cos^2 x + \cos^4 x$?

- 0
- 1
- 2
- 1

If $\sin^2 x + \sin x = 1$

Then $\cos^4 x + \cos^8 x + 2\cos^6 x$?

0

1

2

-1

If $\tan^4 x + \tan^2 x = 1$

Then $\cos^4 x + \cos^2 x$?

0

1

2

-1

If $\sin^2 x + \sin x = 1$

Then $\cos^{12} x + 3\cos^{10} x + 3\cos^8 x + \cos^6 x - 1$?

- 0
- 1
- 2
- 1

If $\operatorname{cosec} x + \operatorname{cosec}^2 x = 1$,

Then $\cot^{12} x + 3 \cot^{10} x + 3 \cot^8 x - \cot^6 x$?

- 0
- 1
- 1
- 2

Sum and Difference of Angles Trigonometric Identities

$$\sin(\alpha+\beta)=\sin(\alpha).\cos(\beta)+\cos(\alpha).\sin(\beta)$$

$$\sin(\alpha-\beta)=\sin\alpha.\cos\beta-\cos\alpha.\sin\beta$$

$$\cos(\alpha+\beta)=\cos\alpha.\cos\beta-\sin\alpha.\sin\beta$$

$$\cos(\alpha-\beta)=\cos\alpha.\cos\beta+\sin\alpha.\sin\beta$$

$$\tan (x+y)=\frac{\tan x+\tan y}{1-\tan x \tan y}$$

$$\tan (x-y)=\frac{\tan x-\tan y}{1+\tan x \tan y}$$

$$\cot (x+y)=\frac{\cot x \cot y-1}{\cot y+\cot x}$$

$$\cot (x-y)=\frac{\cot x \cot y+1}{\cot y-\cot x}$$

SOME EXAMPLES

**Transformations of
sums or differences
into Products**

$$1. \sin C + \sin D = 2 \sin \frac{C+D}{2} \cos \frac{C-D}{2}$$

$$2. \sin C - \sin B = 2 \cos \frac{C+D}{2} \sin \frac{C-D}{2}$$

$$3. \cos C + \cos D = 2 \cos \frac{C+D}{2} \cos \frac{C-D}{2}$$

$$4. \cos C - \cos D = -2 \sin \left(\frac{C+D}{2} \right) \sin \left(\frac{C-D}{2} \right)$$

Trigonometrical Ratios Multiply and Submultiple Angles

If $\frac{1-\tan A}{1+\tan A} = \frac{\tan 7^\circ \times \tan 11^\circ \times \tan 30^\circ \times \tan 79^\circ \times \tan 83^\circ}{\tan 39^\circ \times \tan 41^\circ \times \tan 60^\circ \times \tan 51^\circ \times \tan 49^\circ}$, $\cot^2 A = ?$

18. Find $(1 + \tan 20^\circ)(1 + \tan 24^\circ)(1 + \tan 25^\circ)(1 + \tan 21^\circ)$.

- a) 1
- b) 2
- c) 4
- d) 0

19. Find the value of $\frac{(1+\tan 245^\circ)(1-\tan 200^\circ)}{(1+\tan 250^\circ)(1-\tan 205^\circ)}$.

- a) 2
- b) 1
- c) 0
- d) 4

21. $\tan 13A \cdot \tan 9A \cdot \tan 4A$

- a) $\tan 13A - \tan 9A - \tan 4A$
- b) $\tan 13A + \tan 9A + \tan 4A$
- c) $\tan 13A + \tan 9A - \tan 4A$
- d) $\tan 13A - \tan 9A + \tan 4A$

22. If $\tan 40^\circ = \alpha$, find $\tan 100^\circ$.

a) $\frac{2\alpha}{\alpha^2 - 1}$

b) $\frac{2\alpha}{\alpha^2 + 1}$

c) $\frac{\alpha}{\alpha^2 - 1}$

d) $\frac{\alpha}{\alpha^2 + 1}$

If $\tan \alpha = \frac{1}{7}$, $\sin \beta = \frac{1}{\sqrt{10}}$, where $0 < \alpha < \frac{\pi}{2}$, find $\alpha + 2\beta$.

a. 45

b. 60

c. 90

d. None

Simplify $\frac{2 \cot\left(\frac{\pi-A}{2}\right)}{1+\tan^2\frac{2\pi-A}{2}}$

- a. $\cos A$
- b. $\sin A$
- c. $2 \sin^2 A$
- d. $2 \cos^2 A$

23. If $\tan A = \frac{1}{3}$, $\tan B = \frac{2}{5}$, find $\tan(2A + B)$.

- a) $\frac{14}{23}$
- b) $\frac{23}{14}$
- c) $\frac{12}{13}$
- d) $\frac{5}{7}$

24. $\tan 70^\circ = ?$

- a) $\tan 50^\circ + \tan 20^\circ$
- b) $\tan 50^\circ + 2\tan 20^\circ$
- c) $2\tan 50^\circ + \tan 20^\circ$
- d) $2\tan 50^\circ + 2\tan 20^\circ$

25. If $x + y + z = 180^\circ$, $\tan x \cdot \tan z = 7$, $\tan y \cdot \tan z = 8$,

find $\tan^2 z = ?$

- a) 41
- b) 36
- c) 42
- d) 48

SUPER 100

by *Abbas Cain*

26. $\cot x \cdot \cot 2x - \cot 2x \cot 3x - \cot 3x \cot x =$

- a. 1
- b. 0
- c. -1
- d. *None*

$$(\operatorname{cosec} \theta - \cot \theta)^2 =, 0^\circ < \theta < 90^\circ$$

1. $\frac{1-\cos \theta}{1+\cos \theta}$ 2. $\frac{1+\cos \theta}{1-\sin \theta}$ 3. $\frac{1+\cos \theta}{1-\cos \theta}$ 4. $\frac{1-\sin \theta}{1+\cos \theta}$

The value of $\frac{\sin 4\theta}{(1-\cos 4\theta)}$ is:

1. $\tan \theta$ 2. $\tan 2\theta$ 3. $\cot 2\theta$ 4. $\cot \theta$

Using $2\cos A \cos B = \cos(A + B) + \cos(A - B)$, find the value of $\cos 75^\circ \cos 15^\circ$.

1. $\frac{\sqrt{3}}{4}$ 2. $\frac{\sqrt{3}}{2}$ 3. $\frac{1}{4}$ 4. $\frac{1}{2}$

The value of $\sin 2\theta(\tan \theta + \cot \theta)$ is:

1. 1 2. $\frac{1}{2}$ 3. 2 4. $\frac{2}{3}$

The value of $\frac{\sin 4\theta + \sin 2\theta}{\cos 4\theta + \cos 2\theta}$ is:

1. $\tan 3\theta$ 2. $\tan \theta$ 3. $\cot \theta$ 4. $\cot 3\theta$

Evaluate $\frac{\sin 70^\circ - \sin 50^\circ}{\cos 80^\circ}$.

Evaluate $\cos 80^\circ + \cos 40^\circ - \cos 20^\circ$.

.Evaluate $\sin 70^\circ - \sin 50^\circ - \sin 10^\circ$.

Evaluate $\sin 70^\circ + \cos 40^\circ - \sqrt{3} \sin 80^\circ$.

Evaluate $\frac{\sin 40^\circ + \sin 20^\circ}{\sin 80^\circ}$.

$\frac{\sin A + \sin 3A}{\cos A + \cos 3A}$ is equal to

$\frac{\cos 2B + \cos 2A}{\cos 2B - \cos 2A}$ is equal to

Solve $\frac{\cos 3A - \cos A}{\sin 3A - \sin A} + \frac{\cos 2A - \cos 4A}{\sin 4A - \sin 2A}$.

- a. $\frac{\sin A}{\cos 2A \cos 3A}$
- b. $\frac{\sin 2A}{\cos 2A \cos 3A}$
- c. $\frac{\cos A}{\cos 2A \cos 3A}$
- d. $\frac{\cos 2A}{\cos A \cos 3A}$

$\tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ$

$$\cot 9^\circ - \cot 27^\circ - \cot 63^\circ + \cot 81^\circ$$

5. Find $\frac{\tan 5A + \tan 3A}{4\cos 4A(\tan 5A - \tan 3A)}$.

- a. $\sin 2A$
- b. $\cos 2A$
- c. $\tan 4A$
- d. $\cot 2A$

6. *Find* $\cot(A + 15) - \tan(A - 15)$.

a. $\frac{4\cos 2A}{1+2\sin 2A}$

b. $\frac{4\cos 2A}{1+2\cos 2A}$

c. $\frac{2\cos 2A}{1+2\sin 2A}$

d. $\frac{2\cos 2A}{1+2\cos 2A}$

Ques5. If $A + B = C$, then $\tan A \tan B \tan C = ?$

$\tan C + \tan A + \tan B$

$\tan C - \tan A - \tan B$

$\tan A - \tan B - \tan C$

$\tan C + \tan A - \tan B$

Ques7. Simplify the following:

$$\frac{\cos x - \sqrt{3} \sin x}{2}$$

$$\sin\left(\frac{\pi}{3} - x\right)$$

$$\cos\left(\frac{\pi}{3} + x\right)$$

$$\sin\left(\frac{\pi}{3} + x\right)$$

$$\cos\left(\frac{\pi}{3} - x\right)$$

Ques11. If $\tan 40^\circ = \alpha$, then find $\frac{\tan 320^\circ - \tan 310^\circ}{1 + \tan 320^\circ \tan 310^\circ}$.

$$\frac{1 + \alpha^2}{2\alpha}$$

$$\frac{1 - \alpha^2}{\alpha}$$

$$\frac{1 + \alpha^2}{\alpha}$$

$$\frac{1 - \alpha^2}{2\alpha}$$

**Ques14. If $\tan A - \tan B - \tan C = \tan A \tan B \tan C$,
what is the value of B and C?**

$$A = 2B - 2C$$

$$A = \frac{B - C}{2}$$

$$A = B - C$$

$$A = B + C$$

TRIGONOMETRY

(LECTURE – 4)

If $0 < A < 45^\circ$, $\cos(A + B) = \frac{24}{25}$ & $\sin(A - B) = \frac{15}{17}$,

find $\tan 2A$.

- a. $\frac{213}{4}$
- b. 0
- c. 1
- d. $\frac{416}{87}$

If $\sin(\alpha - \beta) = \frac{5}{13}$, $\cos(\alpha + \beta) = \frac{3}{5}$, find $\tan(2\alpha)$; $\alpha, \beta \in \left(0, \frac{\pi}{4}\right)$

- a. $\frac{63}{16}$
- b. $\frac{61}{16}$
- c. $\frac{65}{16}$
- d. $\frac{32}{9}$

Ques10. If $\tan(\alpha + \beta) = a$, $\tan(\alpha - \beta) = b$, then the value of $\tan 2\alpha$ is:

$$\frac{a-b}{1-ab}$$

$$\frac{a+b}{1-ab}$$

$$\frac{a-b}{1+ab}$$

$$\frac{a+b}{1+ab}$$

If $\cos(\theta + x) = \frac{3}{5}$, $\sin(\theta + y) = \frac{5}{13}$, find $\sin(x - y)$.

- a. $\frac{33}{65}$
- b. $\frac{61}{65}$
- c. $\frac{12}{65}$
- d. $\frac{48}{65}$

10.If $\sin(\theta - x) = a$, $\cos(\theta - y) = b$, find $\cos(x - y)$.

- a) $a\sqrt{1 - b^2} + b\sqrt{1 - a^2}$
- b) $a\sqrt{1 - b^2} - b\sqrt{1 - a^2}$
- c) ab
- d) $2ab$

29. If $\cos x = \frac{11}{61}$ & $\sin y = \frac{4}{5}$. find $\sin^2 \frac{x-y}{2}$.

- a. $\frac{16}{305}$
- b. $\frac{32}{305}$
- c. $\frac{64}{305}$
- d. $\frac{27}{305}$

1. If $(\operatorname{cosec} 2x + \cot 2x)(5 - 2\sqrt{6}) = 1$, then $\sin x = ?$

यदि $(\operatorname{cosec} 2x + \cot 2x)(5 - 2\sqrt{6}) = 1$, तो $\sin x = ?$

A. $\sqrt{\frac{5-2\sqrt{6}}{10}}$

B. $\sqrt{\frac{5-2\sqrt{6}}{5}}$

C. $\sqrt{\frac{5+2\sqrt{6}}{10}}$

D. $\sqrt{\frac{5+2\sqrt{6}}{5}}$

2. If $\operatorname{cosec} \theta - \cot \theta = \frac{1}{3}$, then $(\cos 2\theta + \sin 2\theta) \times \frac{100}{93} = ?$

यदि $\operatorname{cosec} \theta - \cot \theta = \frac{1}{3}$ है तो, $(\cos 2\theta + \sin 2\theta) \times \frac{100}{93} = ?$

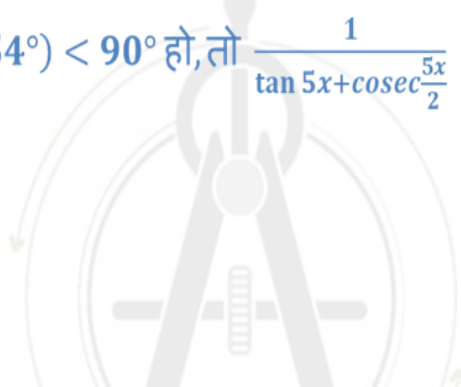
- A. 1.33
- B. 2.66
- C. 8.33
- D. 1.66

THIS BY ABHINAV

If $\cos(2x + 54^\circ) = \sin x$, $0^\circ < (2x + 54^\circ) < 90^\circ$, find $\frac{1}{\tan 5x + \operatorname{cosec} \frac{5x}{2}}$.

यदि $\cos(2x + 54^\circ) = \sin x$, $0^\circ < (2x + 54^\circ) < 90^\circ$ हो, तो $\frac{1}{\tan 5x + \operatorname{cosec} \frac{5x}{2}}$ का मान ज्ञात करें।

- A. $2 + \sqrt{3}$
- B. $3\sqrt{2}$
- C. $2\sqrt{3}$
- D. $2 - \sqrt{3}$



If $\sin 3x = \cos(x + 10^\circ)$, where $3x$ is an acute angle, find $2 \operatorname{cosec} \frac{3x}{2} + 6 \sin^2 3x - \frac{3}{2} \tan^2 3x$

यदि $\sin 3x = \cos(x + 10^\circ)$, जहां $3x$ एक न्यून कोण है, तो $2 \operatorname{cosec} \frac{3x}{2} + 6 \sin^2 3x - \frac{3}{2} \tan^2 3x$ का मान ज्ञात करें।

- A. $7/4$
- B. 4
- C. $17/2$
- D. 5

If $\cos \theta = \frac{1}{2} \left(x + \frac{1}{x} \right)$, then $\frac{1}{2} \left(x^2 + \frac{1}{x^2} \right)$ is.

- a) $\sin 2\theta$
- b) $\cos 2\theta$
- c) $\sec 2\theta$
- d) $\tan 2\theta$

Find $\sqrt{2 + \sqrt{2 + \sqrt{2 + 2\cos 8\theta}}}.$ $\left(0 < \theta < \frac{\pi}{16}\right)$

- a) $2\sin\theta$
- b) $2\cos\theta$
- c) $2\sin 2\theta$
- d) $2\cos 2\theta$

If $\frac{1-\tan A}{1+\tan A} = \frac{\tan 7^\circ \times \tan 11^\circ \times \tan 30^\circ \times \tan 79^\circ \times \tan 83^\circ}{\tan 39^\circ \times \tan 41^\circ \times \tan 60^\circ \times \tan 51^\circ \times \tan 49^\circ},$ $\cot^2 A = ?$

$$\sqrt{\sec^2 \theta + \operatorname{cosec}^2 \theta} \times \sqrt{\tan^2 \theta - \sin^2 \theta} = ?$$

- A. $\sin \theta \cos^2 \theta$
- B. $\sin \theta \sec^2 \theta$
- C. $\operatorname{cosec} \theta \sec^2 \theta$
- D. $\operatorname{cosec} \theta \cos^2 \theta$



If $\operatorname{cosec} \theta = \frac{1}{4x} + x$, then $\operatorname{cosec} \theta + \cot \theta = ?$

यदि $\operatorname{cosec} \theta = \frac{1}{4x} + x$ हो, तो $\operatorname{cosec} \theta + \cot \theta = ?$

- A. $3x$
- B. x
- C. $4x$
- D. $2x$ or $\frac{1}{2x}$



If $\sin 3x \sec 2x = 1$, then $\left[3\tan^2\left(\frac{5x}{2}\right) - 1\right] = ?$

यदि $\sin 3x \sec 2x = 1$ हो, तो $\left[3\tan^2\left(\frac{5x}{2}\right) - 1\right] = ?$

- A. 0
- B. 1
- C. 2
- D. 3

If $\tan \theta + \cot \theta = x$, then $\tan^4 \theta + \cot^4 \theta = ?$

यदि $\tan \theta + \cot \theta = x$ हो, तो $\tan^4 \theta + \cot^4 \theta = ?$

- A. $(x^3 - 3)2 + 2$
- B. $(x^4 - 2x) + 4$
- C. $x(x - 4) + 2$
- D. $x^2(x^2 - 4) + 2$

What is the simplified value of $(1 - \sin A \cos A)(\sin A + \cos A)$?

$(1 - \sin A \cos A)(\sin A + \cos A)$ का सरलीकृत मान क्या है?

- A. $\sin^2 A - \cos^2 A$
- B. $\sin^3 A + \cos^3 A$
- C. 0
- D. $\cos^2 A - \sin^2 A$



If $\cot A = \frac{n}{(n+1)}$ & $\cot B = \frac{1}{(2n+1)}$ then what is the value of $\cot(A + B)$?

यदि $\cot A = \frac{n}{(n+1)}$ व $\cot B = \frac{1}{(2n+1)}$ हो, तो $\cot(A + B)$ का मान क्या है?

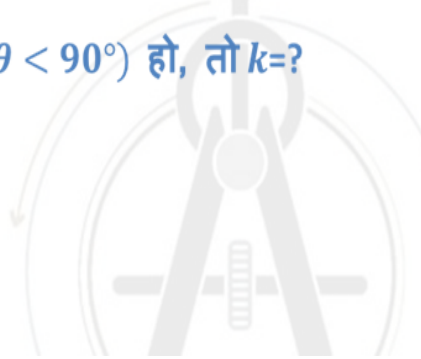
- A. -1
- B. 0
- C. 1
- D. 2



For $0^\circ < \theta < 90^\circ$, if $\frac{\sec \theta (1 - \sin \theta)(\sec \theta + \tan \theta)}{(\sec \theta - \tan \theta)^2} = \frac{1+k}{1-k}$, then $k = ?$

यदि $\frac{\sec \theta (1 - \sin \theta)(\sec \theta + \tan \theta)}{(\sec \theta - \tan \theta)^2} = \frac{1+k}{1-k}$ ($0^\circ < \theta < 90^\circ$) हो, तो $k = ?$

- A. $\operatorname{cosec} \theta$
- B. $\cos \theta$
- C. $\sec \theta$
- D. $\sin \theta$



If $(1 + \tan^2 \theta) + (1 + (\tan^2 \theta)^{-1}) = k$, then $\sqrt{k} = ?$

यदि $(1 + \tan^2 \theta) + (1 + (\tan^2 \theta)^{-1}) = k$ हो, तो $\sqrt{k} = ?$

- A. $\sin \theta \sec \theta$
- B. $\operatorname{cosec} \theta \sec \theta$
- C. $\operatorname{cosec} \theta \cos \theta$
- D. $\sin \theta \cos \theta$



If $117\cos^2 A + 129\sin^2 A = 120$ and $170\cos^2 B + 158\sin^2 B = 161$, then the value of $\operatorname{cosec}^2 A \sec^2 B = ?$
 यदि $117\cos^2 A + 129\sin^2 A = 120$ and $170\cos^2 B + 158\sin^2 B = 161$ हो, तो $\operatorname{cosec}^2 A \sec^2 B$ का मान क्या है?

- A. 16
- B. 4
- C. 1
- D. 9



If $\tan^4 x - \tan^2 x = 1$, then the value of $\sin^4 x + \sin^2 x$ is:
 यदि $\tan^4 x - \tan^2 x = 1$ हो तो $\sin^4 x + \sin^2 x$ का मान क्या है?

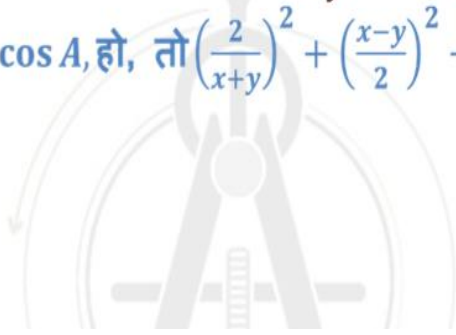
- A. 3/4
- B. 1
- C. 3/2
- D. 1/2



If $x = \operatorname{cosec} A + \cos A$ & $y = \operatorname{cosec} A - \cos A$, then $\left(\frac{2}{x+y}\right)^2 + \left(\frac{x-y}{2}\right)^2 - 1 = ?$

यदि $x = \operatorname{cosec} A + \cos A$ & $y = \operatorname{cosec} A - \cos A$ हो, तो $\left(\frac{2}{x+y}\right)^2 + \left(\frac{x-y}{2}\right)^2 - 1 = ?$

- A. 2
- B. 1
- C. 0
- D. 3



If $\sec \theta + \tan \theta = 2 + \sqrt{5}$ and θ is an acute angle, find the value of $\sin \theta$.

यदि $\sec \theta + \tan \theta = 2 + \sqrt{5}$ हो और θ एक न्यून कोण है तो का $\sin \theta$ मान क्या है?

- A. $\frac{2\sqrt{5}}{5}$
- B. $\frac{1}{5}$
- C. $\frac{\sqrt{5}}{5}$
- D. $\frac{3}{5}$



1. $3 - 2 \sin^2 \theta - 3 \cos \theta = 0$, $0 < \theta < 90^\circ$, find θ .

2. $4 - 2 \sin^2 A - 5 \cos A = 0$, $0 < A < 90^\circ$, find A .

3. $7 \sin^2 A - \cos^2 A + 2 \sin A = 2$, $0 < A < 90^\circ$, find A .

4. $12 \cot^2 A - 31 \operatorname{cosec} A + 32 = 0$, find $\tan A$.

If $4\sin^2 \theta = 3(1 + \cos \theta)$, $0^\circ < \theta < 90^\circ$, then what is the value of $(2\tan \theta + 4\sin \theta - \sec \theta)$?

$$3\sqrt{15} - 4$$

$$15\sqrt{3} - 4$$

$$15\sqrt{3} + 3$$

$$4\sqrt{15} - 3$$

If $\frac{\sin^2 \emptyset - 3\sin \emptyset + 2}{\cos^2 \emptyset} = 1$, where $0^\circ < \emptyset < 90^\circ$, then what is the value of $(\cos 2\emptyset - \sin 3\emptyset + \operatorname{cosec} 2\emptyset)$?

$$\frac{-3 - 4\sqrt{3}}{6}$$

$$\frac{-3 + 4\sqrt{3}}{6}$$

$$\frac{3 + 4\sqrt{3}}{6}$$

$$\frac{3 - 4\sqrt{3}}{6}$$

If $\tan^2 A - 6 \tan A + 9 = 0$, $0 < A < 90^\circ$, What is the value of $6 \cot A + 8\sqrt{10}\cos A$?

$$10\sqrt{10}$$

$$14$$

$$10$$

$$8\sqrt{10}$$

If $6 \tan A (\tan A + 1) = 5 - \tan A$, Given that $0 < A < \frac{\pi}{2}$ what is the value of $(\sin A + \cos A)$?

$$3\sqrt{5}$$

$$\frac{5}{\sqrt{3}}$$

$$5\sqrt{3}$$

$$\frac{3}{\sqrt{5}}$$

If $3 \sin^2 \theta + 4 \cos \theta - 4 = 0$, $0^\circ < \theta < 90^\circ$, then the value of $(\operatorname{cosec}^2 \theta + \cot^2 \theta)$ is

$$\frac{5}{4}$$

$$\frac{25}{3}$$

$$\frac{4}{3}$$

$$\frac{17}{9}$$

If $2 \sin^2 \theta + 3 \cos \theta = 3$, $0^\circ < \theta < 90^\circ$, then the value of $(\sec^2 \theta + \cot^2 \theta)$ is

$$3\frac{2}{3}$$

$$3\frac{1}{3}$$

$$4\frac{1}{3}$$

$$4\frac{1}{2}$$

If $\sec^2 \theta + \tan^2 \theta = 3\frac{1}{2}$, $0^\circ < \theta < 90^\circ$, then $(\cos\theta + \sin\theta)$ is equal to

$$\frac{1+\sqrt{5}}{3}$$

$$\frac{2+\sqrt{5}}{3}$$

$$\frac{1+\sqrt{5}}{6}$$

$$\frac{9+2\sqrt{5}}{6}$$

If $\tan^2 A + 2\tan A - 63 = 0$ Given that $0 < A < \frac{\pi}{2}$ what is the value of $(2\sin A + 5\cos A)$?

$$19\sqrt{50}$$

$$15\sqrt{50}$$

$$\frac{19}{\sqrt{50}}$$

$$\frac{15}{\sqrt{50}}$$

Thankyou

TRIGONOMETRY

(LECTURE – 5)

Find the smallest positive angle which satisfies the given trigonometric equation:

सबसे छोटा धनात्मक कोण ज्ञात कीजिए जो दिए गए त्रिकोणमितीय समीकरण को संतुष्ट करता है

$$2\sin^2 x + \sqrt{3} \cos x + 1 = 0$$

A. $\frac{\pi}{6}$

B. $\frac{2\pi}{3}$

C. $\frac{5\pi}{3}$

D. $\frac{\pi}{3}$



If $\sin \theta \cdot \sec^2 \theta = \frac{2}{3}$ ($0^\circ < \theta < 90^\circ$), then $(\tan^2 \theta + \cos^2 \theta) = ?$

यदि $\sin \theta \cdot \sec^2 \theta = \frac{2}{3}$ ($0^\circ < \theta < 90^\circ$) हो, तो $(\tan^2 \theta + \cos^2 \theta) = ?$

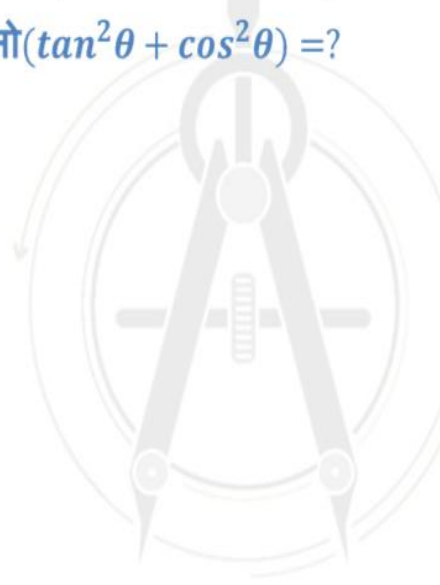
SSC CHSL July 3 2019 (morning)

A. $\frac{7}{6}$

B. $\frac{11}{12}$

C. $\frac{13}{12}$

D. $\frac{5}{4}$



If $\cot^4 \theta + \cos^2 \theta = 1$ and $\theta < 90^\circ$, find the value of $\operatorname{cosec}^3 \theta (\sin^2 \theta - 1)$.

यदि $\cot^4 \theta + \cos^2 \theta = 1$ व $\theta < 90^\circ$, हो तो $\operatorname{cosec}^3 \theta (\sin^2 \theta - 1)$ का मान खोजे।

A. -1

B. 0

C. 1

D. -2



14. $\sin(90^\circ + 2A) [4 - \cos^2(90^\circ - 2A)]$

- a. $2(\cos^3 A - \sin^3 A)$
- b. $2(\cos^3 A + \sin^3 A)$
- c. $4(\cos^6 A + \sin^6 A)$
- d. $4(\cos^6 A - \sin^6 A)$

20. $\sin(630^\circ + A) + \cos A =$

- a. 0
- b. 1
- c. -1
- d. 2

21. $\frac{2 \sin(45^\circ + \theta) \sin(45^\circ - \theta)}{\cos 2\theta} =$

- a. 0
- b. $\tan 2\theta$
- c. $\cot 2\theta$
- d. 1

22. $\frac{\cos(90+A)}{\sec(270-A)} + \frac{\sin(270+A)}{\operatorname{cosec}(630-A)} =$

- a. 0
- b. 1
- c. -1
- d. 2

23. $\frac{[\tan(90-\theta)+\cot(90-\theta)]^2}{2 \sec^2(90-2\theta)} =$

- a. 1
- b. 2
- c. -1
- d. 0

24. $\frac{[\sin(90-A)+\cos(180-2A)]}{[\cos(90-2A)+\sin(180-A)]} =$

- a. $\sin\left(\frac{A}{2}\right) \cdot \cos A$
- b. $\cot\left(\frac{A}{2}\right)$
- c. $\tan\left(\frac{A}{2}\right)$
- d. $\sin A \cdot \cos\left(\frac{A}{2}\right)$

If $\cos^2 \theta + 3 = 3(\cot^2 \theta + \sin^2 \theta)$, $0 < \theta < 90^\circ$, then what is the value of $(2\cos \theta + 3\sin \theta)$?

यदि $\cos^2 \theta + 3 = 3(\cot^2 \theta + \sin^2 \theta)$, $0 < \theta < 90^\circ$, तो $(2\cos \theta + 3\sin \theta)$ का मान क्या है?

$$\frac{1+3\sqrt{3}}{2}$$

$$\frac{2\sqrt{3}+1}{2}$$

$$\frac{\sqrt{3}+2}{2}$$

$$\frac{2+3\sqrt{3}}{2}$$

The expression $\frac{(1 + \cos \theta + \sin \theta) \div (1 + \cos \theta - \sin \theta)}{(\sec \theta - \cos \theta)(\cot \theta + \tan \theta)(1 + \sin \theta)}$

, $0^\circ < \theta < 90^\circ$, is equal to:

व्यंजक $\frac{(1 + \cos \theta + \sin \theta) \div (1 + \cos \theta - \sin \theta)}{(\sec \theta - \cos \theta)(\cot \theta + \tan \theta)(1 + \sin \theta)}$ $0^\circ < \theta < 90^\circ$ के बराबर है:

- $\tan \theta$
- $\cot \theta$
- $\sec \theta$
- $\sin \theta$

$$(\tan^2 \theta + \cot^2 \theta + 2) \div \left(\frac{1}{1+\cos \theta} + \frac{1}{1-\cos \theta} \right),$$

$0^\circ < \theta < 90^\circ$, is equal to/ **के बराबर है:**

- $(\operatorname{cosec}^2 \theta)/2$
- $2\operatorname{cosec}^2 \theta$
- $(\sec^2 \theta)/2$
- $2\sec^2 \theta$

If $\sec \theta + \tan \theta = k$, $0^\circ < \theta < 90^\circ$, then $\frac{(k-1)^2 - 2}{(k+1)^2 - 2k}$

is equal to:

यदि $\sec \theta + \tan \theta = k$, $0^\circ < \theta < 90^\circ$, तो $\frac{(k-1)^2 - 2}{(k+1)^2 - 2k}$

बराबर है:

- $\sin \theta - \cos \theta$
- $\cos \theta + \sin \theta$
- $\sec \theta \operatorname{cosec} \theta$
- $\cos \theta \sin \theta$

$$\frac{(\sin\theta - \cos\theta)(1 + \tan\theta + \cot\theta)(\operatorname{cosec}^2\theta)}{\sec^3\theta - \operatorname{cosec}^3\theta}, 0^\circ < \theta < 90^\circ,$$

is equal to / के बराबर है:

- $\sec\theta \operatorname{cosec}\theta$
- $\cos^2\theta$
- $\sin^2\theta$
- $\sin\theta \cos\theta$

$$\frac{\tan\theta + \sec\theta - 1}{\tan\theta - \sec\theta + 1} \div \left(1 + \frac{\tan^2\theta}{1 + \sec\theta}\right)$$

is equal to / बराबर है $0^\circ < \theta < 90^\circ$

- $\sin\theta + \cos\theta$
- $1 + \cos\theta$
- $1 + \sin\theta$
- $\sin\theta \cos\theta$

The value of $\left(\frac{1-\cot\theta}{1-\tan\theta}\right)^2 + 1$, if $0^\circ < \theta < 90^\circ$, is equal to:

$\operatorname{cosec}^2\theta$

$\sin^2\theta$

$\cos^2\theta$

$\sec^2\theta$

Simplify the following expression:

$$\operatorname{cosec}^4 A (1 - \cos^4 A) - 2 \cot^2 A - 1$$

$\sin^2 A$

$\operatorname{cosec}^2 A$

1

0

Simplify the following expression

$$\frac{\cos A}{1 - \tan A} + \frac{\sin A}{1 - \cot A} - \sin A$$

$$1 + \cos A$$

$$(1 + \sin A) \cos A$$

$$1 + \sin A$$

$$\cos A$$

If $\cos 53^\circ = \frac{x}{y}$, then $\sec 53^\circ + \cot 37^\circ$ is equal to:

$$\frac{x + \sqrt{y^2 - x^2}}{y}$$

$$\frac{x + \sqrt{y^2 - x^2}}{x}$$

$$\frac{y + \sqrt{y^2 - x^2}}{x}$$

$$\frac{y + \sqrt{y^2 - x^2}}{y}$$

The value of $2 - \sqrt{\frac{\cot\theta + \cos\theta}{\cot\theta - \cos\theta}}$, when $0^\circ < \theta < 90^\circ$ is equal to:

$2 + \sec\theta + \tan\theta$

$2 - \sec\theta + \tan\theta$

$2 - \sec\theta - \tan\theta$

$2 + \sec\theta - \tan\theta$

The expression $(\cos^6\theta + \sin^6\theta - 1)(\tan^2\theta + \cot^2\theta + 2) + 3$ is equal to:

0

1

2

-1

The value of $\left(\frac{1-\cot\theta}{1-\tan\theta}\right)^2 - 1$ when $0^\circ < \theta < 90^\circ$, is equal to:

$$\cos^2 \theta - 1$$

$$\sec^2 \theta + 1$$

$$\cot^2 \theta - 1$$

$$\sin^2 \theta - 1$$

The value of $1 + \sqrt{\frac{\cot\theta + \cos\theta}{\cot\theta - \cos\theta}}$, if $0^\circ < \theta < 90^\circ$, is equal to:

$$1 - \sec\theta + \tan\theta$$

$$1 - \sec\theta - \tan\theta$$

$$1 + \sec\theta - \tan\theta$$

$$1 + \sec\theta + \tan\theta$$

The value of $\sin 73^\circ + \cos 137^\circ$ is:

$\cos 13^\circ$

$\sin 13^\circ$

$\cos 18^\circ$

$\sin 18^\circ$

$\frac{1+\cos \theta - \sin^2 \theta}{\sin \theta (1+\cos \theta)} \times \frac{\sqrt{\sec^2 \theta + \operatorname{cosec}^2 \theta}}{\tan \theta + \cot \theta}$, $0^\circ < \theta < 90^\circ$, is equal to

$\frac{1+\cos \theta - \sin^2 \theta}{\sin \theta (1+\cos \theta)} \times \frac{\sqrt{\sec^2 \theta + \operatorname{cosec}^2 \theta}}{\tan \theta + \cot \theta}$, $0^\circ < \theta < 90^\circ$ का मान इसमें से किसके बराबर है?

(a) $\sec \theta$

(b) $\tan \theta$

(c) $\operatorname{cosec} \theta$

(d) $\cot \theta$

$$\left(\frac{\tan^3 \theta}{\sec^2} + \frac{\cot^3 \theta}{\operatorname{cosec}^2} + 2 \sin \theta \cos \theta \right) \div (1 + \operatorname{cosec}^2 \theta + \tan^2 \theta), 0^\circ < \theta < 90^\circ$$

is equal to:

$$\left(\frac{\tan^3 \theta}{\sec^2} + \frac{\cot^3 \theta}{\operatorname{cosec}^2} + 2 \sin \theta \cos \theta \right) \div (1 + \operatorname{cosec}^2 \theta + \tan^2 \theta), 0^\circ < \theta < 90^\circ$$

का मान इनमें से किसके बराबर होगा?

Cosecx

Secx

Sinx cosx

Cosecx cosx

Let $0^\circ < \theta < 90^\circ$. $(1 + \cot^2 \theta) (1 + \tan^2 \theta) \times (\sin \theta - \operatorname{cosec} \theta)(\cos \theta - \sec \theta)$ is equal to:

मान लें कि $0^\circ < \theta < 90^\circ$ है। तो $(1 + \cot^2 \theta) (1 + \tan^2 \theta) \times (\sin \theta - \operatorname{cosec} \theta)(\cos \theta - \sec \theta)$ का मान इनमें से किसके बराबर होगा?

(a) $\sec \theta + \operatorname{cosec} \theta$

(b) $\sin \theta \cos \theta$

(c) $\sin \theta + \cos \theta$

(d) $\sec \theta \operatorname{cosec} \theta$

The expression $(\tan \theta + \cot \theta)(\sec \theta + \tan \theta)(1 - \sin \theta)$, $0^\circ < \theta < 90^\circ$, is equal to:

$\sec \theta$

$\operatorname{cosec} \theta$

$\cot \theta$

$\sin \theta$

The expression $\frac{\tan^6 \theta - \sec^6 \theta + 3 \sec^2 \theta \tan^2 \theta}{\tan^2 \theta + \cot^2 \theta + 2}$, $0^\circ < \theta < 90^\circ$, is equal to:

$\sec^2 \theta \operatorname{cosec}^2 \theta$

$-\sec^2 \theta \operatorname{cosec}^2 \theta$

$\cos^2 \theta \sin^2 \theta$

$-\cos^2 \theta \sin^2 \theta$

The expression $\frac{\cos^4 \theta - \sin^4 \theta + 2 \sin^2 \theta + 3}{(\operatorname{cosec} \theta + \cot \theta + 1)(\operatorname{cosec} \theta - \cot \theta + 1) - 2}$, $0^\circ < \theta < 90^\circ$, is equal to:

$$\frac{1}{2} \sin \theta$$

$$2 \sin \theta$$

$$\sec \theta$$

$$2 \operatorname{cosec} \theta$$

The expression $\frac{(1 - 2 \sin^2 \theta \cos^2 \theta)(\cot \theta + 1) \cos \theta}{(\sin^4 \theta + \cos^4 \theta)(1 + \tan \theta) \operatorname{cosec} \theta} - 1$, $0^\circ < \theta < 90^\circ$, equals:

$$\cos^2 \theta$$

$$-\sin^2 \theta$$

$$\sec^2 \theta$$

$$-\sec^2 \theta$$

The expression $\frac{(1-\sin \theta+\cos \theta)^2(1-\cos \theta) \sec ^3 \theta \operatorname{cosec}^2 \theta}{(\sec \theta-\tan \theta)(\tan \theta+\cot \theta)}, 0^{\circ} < \theta < 90^{\circ}$, is equal to:

$$\sin \theta$$

$$2 \cos \theta$$

$$\cot \theta$$

$$2 \tan \theta$$

Solve the given equation.

$$\sin 21^{\circ} \cos 9^{\circ}-\cos 84^{\circ} \cos 6^{\circ}=?$$

$$1$$

$$\frac{1}{2}$$

$$\frac{1}{4}$$

$$\frac{1}{8}$$

$\sin^6 X + \cos^6 X = 1/3, 0^\circ < x < 90^\circ$, find $\sin X \cos X$?

$\sqrt{2}/\sqrt{3}$

$\sqrt{2}/3$

$2/3$

$2/\sqrt{3}$

9. If $\sin x + \sin y = 3(\cos y - \cos x)$, find $\frac{\sin 3x}{\sin 3y}$

- a. 1
- b. -1
- c. 0
- d. *CND*

Ques1. If $2 \frac{\cos^2 x - \sec^2 x}{\tan^2 x} = a + b \cos 2x$, then a, b = ?

$$\frac{3}{2}, \frac{1}{2}$$

$$-\frac{3}{2}, -\frac{1}{2}$$

$$3, 1$$

$$-3, -1$$

Ques2. If $\tan A \tan B + \frac{\cos x}{\cos A \cos B} = 1$, then x = ?

$$\bullet B$$

$$\bullet A$$

$$\bullet A - B$$

$$\bullet A + B$$

Ques3. What is the value of the expression?
 $\cos 2A \cos 2B + \sin^2(A - B) - \sin^2(A + B)$

$\cos(2A - 2B)$

$\sin(2A + 2B)$

$\sin(2A - 2B)$

$\cos(2A + 2B)$

Ques4. If $x = 8(\sin\theta + \cos\theta)$ and $y = 9(\sin\theta - \cos\theta)$, then the value of $\frac{x^2}{8^2} + \frac{y^2}{9^2}$ is:

•6

•2

•8

•4

**Ques6. $\sin^4\theta + \cos^4\theta$,
in terms of $\sin\theta$ can be written as: –**

$$2\sin^4\theta - 2\sin^2\theta + 1$$

$$2\sin^4\theta + 2\sin^2\theta + 1$$

$$2\sin^4\theta - 2\sin^2\theta$$

$$2\sin^4\theta - 2\sin^2\theta - 1$$

**Ques12. If $\sin^2\theta\cos^2\theta = \frac{2}{9}$,
then what will be the value of $\operatorname{cosec}^2\theta + \sec^2\theta$?**

$$9\sqrt{2}$$

$$\frac{9}{2}$$

$$\frac{5}{2}$$

$$\frac{7}{2}$$

**Ques13. If $1 + \sin\theta = m\cos\theta$,
then what is the value of $\sin\theta$?**

$$\frac{m^2+1}{m^2-1}$$

$$\frac{m^2-1}{m^2+1}$$

$$\frac{2m^2+1}{m^2-1}$$

$$\frac{m^2+1}{2m^2-1}$$

If $a = \frac{2\sin x}{1+\sin x+\cos x}$, $b = \frac{c}{1+\sin x}$, if $a = b$, find c .

a) $1 + \sin x - \cos x$

b) $1 - \sin x \cdot \cos x$

c) $1 + \sin x \cdot \cos x$

d) $1 + \cos x - \sin x$

$$\text{if } a = (\cot^2 x - 1) / (1 + \cot x + \operatorname{cosec} x)$$

$$b = C \cdot \cot 2x$$

If $a = b$ Find C

$$(1 + \cot x - \operatorname{cosec} x)$$

$$(1 + \cot x + \operatorname{cosec} x)$$

$$(1 - \cot x - \operatorname{cosec} x)$$

$$(1 - \cot x + \operatorname{cosec} x)$$

$$\text{If } a = \cos X / (\tan X - \sec X + 1)$$

$$b = (1 + \sin X) / C$$

Find C if $a = b$

$$\tan X + \sec X - 1$$

$$\tan X + \sec X + 1$$

$$\tan X - \sec X - 1$$

$$\tan X - \sec X + 1$$

If $\cos^2 \alpha - \sin^2 \alpha = \tan^2 \beta$, find $\cos^2 \beta - \sin^2 \beta$

- a. $\tan^2 \alpha$
- b. $\cot^2 \alpha$
- c. $\cot^2 \beta$
- d. $\tan^2 \beta$

If $\frac{\cos \alpha}{\sin \alpha + \cos \beta} + \frac{\cos \beta}{\sin \beta - \cos \alpha} = \frac{x}{\sin \alpha - \cos \beta} + \frac{\cos \beta}{\sin \beta + \cos \alpha}$ then $x = ?$

यदि $\frac{\cos \alpha}{\sin \alpha + \cos \beta} + \frac{\cos \beta}{\sin \beta - \cos \alpha} = \frac{x}{\sin \alpha - \cos \beta} + \frac{\cos \beta}{\sin \beta + \cos \alpha}$ हो, तो $x = ?$

- A. $\cos \beta$
- B. $\cos \alpha$
- C. $\sin \beta$
- D. $\sin \alpha$

Thankyou

TRIGONOMETRY

(LECTURE – 6)

$\sin 10^\circ - \frac{4}{3} (\sin^3 10^\circ)$?

- A. $\frac{1}{2}$
- B. $\frac{1}{3}$
- C. $\frac{1}{5}$
- D. $\frac{1}{6}$

$$4/3(\cos^3 20^\circ) - \cos 20^\circ ?$$

A. $1/2$

B. $1/3$

C. $1/5$

D. $1/6$

$$\sin x + \cos x = (\sqrt{3}-1)/2\sqrt{2}$$

Find $\tan x + \cos x$?

$$28.32 \cos^6 x - 48 \cos^4 x + 18 \cos^2 x - 1 =$$

- a. $\cos 3x$
- b. $\cos 6x$
- c. $\sin 6x$
- d. $\sin 3x$

$$\sin 12^\circ \cdot \sin 48^\circ \cdot \sin 54^\circ =$$

- a. $\frac{1}{2}$
- b. $\frac{1}{4}$
- c. $\frac{1}{8}$
- d. $\frac{1}{16}$

$$\sin 20^\circ \cdot \sin 100^\circ \cdot \sin 140^\circ \cdot \sin 60^\circ =$$

- a. $\frac{1}{8}$
- b. $\frac{3}{16}$
- c. $\frac{1}{32}$
- d. $\frac{1}{16}$

$$\cos\left(\frac{\pi}{15}\right) \cdot \cos\left(\frac{2\pi}{15}\right) \cdot \cos\left(\frac{3\pi}{15}\right) \cdot \cos\left(\frac{4\pi}{15}\right) \cdot \cos\left(\frac{5\pi}{15}\right) \cdot \cos\left(\frac{6\pi}{15}\right) \cdot \cos\left(\frac{7\pi}{15}\right) =$$

- a. $\frac{1}{32}$
- b. $\frac{1}{64}$
- c. $\frac{1}{128}$
- d. $\frac{1}{256}$

$$\sin\left(\frac{\pi}{14}\right) \cdot \sin\left(\frac{3\pi}{14}\right) \cdot \sin\left(\frac{5\pi}{14}\right) \cdot \sin\left(\frac{7\pi}{14}\right) \cdot \sin\left(\frac{9\pi}{14}\right) \cdot \sin\left(\frac{11\pi}{14}\right) \cdot \sin\left(\frac{13\pi}{14}\right) =$$

- a. $\frac{1}{16}$
- b. $\frac{1}{32}$
- c. $\frac{1}{64}$
- d. $\frac{1}{128}$

The value of $\frac{\sin 23^\circ \cos 67^\circ + \sec 52^\circ \sin 38^\circ + \cos 23^\circ \sin 67^\circ + \operatorname{cosec} 52^\circ \cos 38^\circ}{\operatorname{cosec}^2 20^\circ - \tan^2 70^\circ}$ is:

- 3
- 4
- 2
- 0

The value of $\frac{3(\cot^2 47^\circ - \sec^2 43^\circ) - 2(\tan^2 23^\circ - \operatorname{cosec}^2 67^\circ)}{\operatorname{cosec}^2 (68^\circ + \theta) - \tan(\theta + 61^\circ) - \tan^2 (22^\circ - \theta) + \cot(29^\circ - \theta)}$ is:

-1

1

5

0

The value of $\frac{3(\operatorname{cosec}^2 26^\circ - \tan^2 64^\circ) + (\cot^2 42^\circ - \sec^2 48^\circ)}{\cot(22^\circ - \theta) - \operatorname{cosec}^2 (62^\circ + \theta) - \tan(\theta + 68^\circ) + \tan^2 (28^\circ - \theta)}$ is

3

4

-1

-2

The value of $\frac{(\cos 9^\circ + \sin 81^\circ)(\sec 9^\circ + \operatorname{cosec} 81^\circ)}{\operatorname{cosec}^2 71^\circ + \cos^2 15^\circ - \tan^2 19^\circ + \cos^2 75^\circ}$ is:

- 1
- 4
- 3
- 2

The value of $\frac{4 \tan^2 30^\circ + \sin^2 30^\circ \cos^2 45^\circ + \sec^2 48^\circ - \cot^2 42^\circ}{\cos 37^\circ \sin 53^\circ + \sin 37^\circ \cos 53^\circ + \tan 18^\circ \tan 72^\circ}$ is:

- $\frac{35}{48}$
- $\frac{59}{48}$
- $\frac{49}{24}$
- $\frac{35}{24}$

$$\frac{\operatorname{cosec}(78^\circ + \theta) - \sec(12^\circ - \theta) - \tan(67^\circ + \theta) + \cot(23^\circ - \theta)}{\tan 13^\circ \tan 37^\circ \tan 45^\circ \tan 53^\circ \tan 77^\circ} = ?$$

SSC CGL 5-3-2020 Afternoon

- A. 0
- B. 2
- C. -1
- D. 1



$$\frac{(\cos 9^\circ + \sin 81^\circ)(\sec 9^\circ + \operatorname{cosec} 81^\circ)}{\sin 56^\circ \sec 34^\circ + \cos 25^\circ \operatorname{cosec} 65^\circ} = ?$$

SSC CHSL 2-7-2019 Evening

- A. 1/2
- B. 4
- C. 2
- D. 1

If $\operatorname{cosec} 31^\circ = x$, then $\sin^2 59 + \frac{1}{\operatorname{cosec}^2 31} + \tan^2 59 - \frac{1}{\sin^2 59 \cdot \operatorname{cosec}^2 59} = ?$

यदि $\operatorname{cosec} 31^\circ = x$ हो, तो $\sin^2 59 + \frac{1}{\operatorname{cosec}^2 31} + \tan^2 59 - \frac{1}{\sin^2 59 \cdot \operatorname{cosec}^2 59} = ?$

SSC CHSL 2-7-2019 Afternoon

A. $x + 1$

B. $x^2 - 1$

C. $x + 1$

D. $x^2 + 1$



13. Find $\sin 10^\circ + \sin 20^\circ + \sin 30^\circ + \dots + \sin 350^\circ + \sin 360^\circ$.

a) 0

b) 1

c) $\sin 10^\circ - \sin 360^\circ$

d) $\sin 10^\circ + \sin 360^\circ$

Find $\cos 10^\circ + \cos 20^\circ + \cos 30^\circ + \dots + \cos 360^\circ$.

- a) 0
- b) 1
- c) $\sin 10^\circ - \sin 360^\circ$
- d) $\sin 10^\circ + \sin 360^\circ$

Find $\cos^2 5^\circ + \cos^2 10^\circ + \cos^2 15^\circ + \dots + \cos^2 90^\circ$.

$\sin^2 5^\circ + \sin^2 10^\circ + \dots + \sin^2 90^\circ = ?$

.Find $\cos\theta.\cos2\theta.\cos4\theta.\cos8\theta \dots \dots \cos2^n\theta$.

- a. $\frac{\sin2^{n+1}\theta}{2^{n+1}\sin\theta}$
- b. $\frac{\sin2^{n+1}\theta}{2^n\sin\theta}$
- c. $\frac{\sin2^n\theta}{2^n\sin\theta}$
- d. *None*

$$\frac{\cos3^\circ}{\sin1^\circ.\cos2^\circ} + \frac{\cos5^\circ}{\sin2^\circ.\cos3^\circ} + \dots + \frac{\cos177^\circ}{\sin88^\circ.\cos89^\circ} =$$

- a. 0
- b. 1
- c. $\cos3^\circ - 1$
- d. $1 - \cos89^\circ$

12. If $\cos x + \sin x = \sqrt{2} \cos x$, what is the value of $(\cos x - \sin x)^2 + (\cos x + \sin x)^2$?

यदि $\cos x + \sin x = \sqrt{2} \cos x$ है, $(\cos x - \sin x)^2 + (\cos x + \sin x)^2$ का मान क्या है?

a) 2

b) 1

c) 0

d) $\frac{1}{\sqrt{2}}$

17. Using the identity $\tan 2\alpha = \frac{2\tan \alpha}{1 - \tan^2 \alpha}$, find the value of $\tan 15^\circ$, correct to three decimal places. [use $\sqrt{3} = 1.732$]

पहचान $\tan 2\alpha = \frac{2\tan \alpha}{1 - \tan^2 \alpha}$ का उपयोग करके, $\tan 15^\circ$ का मान ज्ञात कीजिये, जो तीन दशमलव स्थानों से सही है। [उपयोग $\sqrt{3} = 1.732$]

a) 0.268

b) 0.27

c) 0.267

d) 0.269

19. Using $\operatorname{cosec}(\alpha + \beta) = \frac{\sec \alpha \times \sec \beta \times \operatorname{cosec} \alpha \times \operatorname{cosec} \beta}{\sec \alpha \times \operatorname{cosec} \beta + \operatorname{cosec} \alpha \times \sec \beta}$, find the value of $\operatorname{cosec} 75^\circ$.

$$\operatorname{cosec}(\alpha + \beta) = \frac{\sec \alpha \times \sec \beta \times \operatorname{cosec} \alpha \times \operatorname{cosec} \beta}{\sec \alpha \times \operatorname{cosec} \beta + \operatorname{cosec} \alpha \times \sec \beta}$$

का उपयोग करके, $\operatorname{cosec} 75^\circ$ का मान ज्ञात कीजिये।

a) $\frac{\sqrt{6} + \sqrt{2}}{4}$

b) $\frac{\sqrt{6} - \sqrt{2}}{4}$

c) $\sqrt{6} - \sqrt{2}$

d) $\sqrt{6} + \sqrt{2}$

If $x = 1 + \operatorname{cosec}^2 \theta + \tan^2 \theta$, $y = (\tan \theta + \cot \theta) \sin \theta$ and $z = \frac{\tan^2 \theta}{\sec^2 \theta} + \frac{\cot^2 \theta}{\operatorname{cosec}^2 \theta} + 2 \sin \theta \cos \theta$, $0^\circ < \theta < 90^\circ$, then $(x \div z \times y)$ is equal to:

यदि $x = 1 + \operatorname{cosec}^2 \theta + \tan^2 \theta$, $y = (\tan \theta + \cot \theta) \sin \theta$ और $z = \frac{\tan^2 \theta}{\sec^2 \theta} + \frac{\cot^2 \theta}{\operatorname{cosec}^2 \theta} + 2 \sin \theta \cos \theta$, $0^\circ < \theta < 90^\circ$, तो $(x \div z \times y)$ किसके बराबर है?

a) $\sec^2 \theta \operatorname{cosec}^2 \theta$

b) $\sec^2 \theta \operatorname{cosec} \theta$

c) $\sec \theta \operatorname{cosec}^2 \theta$

d) $\sec \theta \operatorname{cosec} \theta$

If $\sin\theta + \tan\theta = p$ and $\tan\theta - \sin\theta = q$, then which of the relation satisfies for given values of p and q ?

1) $p^2 + q^2 = 4pq$

2) $p^2 - q^2 = 4\sqrt{pq}$

3) $p^2 + q^2 = 4\sqrt{pq}$

4) $p^2 - q^2 = 8\sqrt{pq}$

If $\cot A + \cos A = p$ and $\cot A - \cos A = q$, then which of the following relation is correct?

(A) $\sqrt{\frac{1}{16pq}} = p^2 + q^2$

(B) $\frac{1}{4pq} = p^2 + q^2$

(C) $\sqrt{16pq} = p^2 - q^2$

(D) $4pq = p^2 + q^2$

What is the value of the expression

$$\sin A \left(1 + \frac{\sin A}{\cos A} \right) + \cos A \left(1 + \frac{\cos A}{\sin A} \right) ?$$

$$\sec A + \operatorname{cosec} A$$

$$\sin A + \cos A$$

$$\sin A - \cos A$$

$$\sec A - \operatorname{cosec} A$$

Simplify the expression:

$$\frac{3 - \sin^2 A + \cos^2 A}{2 + 2 \cos^2 A}$$

1

0

-1

2

$$(1 + \tan^2 A)(1 + \cot^2 A) = ?$$

$$\frac{1}{\cos^2 A(1 + \sin^2 A)}$$

$$\frac{1}{\sin^2 A(1 - \sin^2 A)}$$

$$\frac{1}{\sin^2 A + \operatorname{cosec}^2 A}$$

$$\frac{1}{\sin^2 A(1 + \cos^2 A)}$$

If $\operatorname{cosec} A + \cot A = a\sqrt{b}$, then find the value of $\frac{(a^2 b - 1)}{(a^2 b + 1)}$

$$\cos A$$

$$\tan A$$

$$\frac{1}{\sin A}$$

$$\frac{1}{\cot A}$$

If $\cos \theta + \sec \theta = \sqrt{3}$, then the value of $\cos^3 \theta + \sec^3 \theta$ is:

0

$\frac{1}{\sqrt{3}}$

$\sqrt{3}$

$2\sqrt{3}$

The value of the expression $4(\sin^6 A + \cos^6 A) - 6(\sin^4 A + \cos^4 A) + 8$ is :

4

8

7

6

What is the value of $\cos 1^\circ \cdot \cos 2^\circ \cdot \cos 3^\circ \dots \cos 177^\circ \cdot \cos 178^\circ \cdot \cos 179^\circ$?

0

1

$1/\sqrt{2}$

$1/2$

The solution of the equation $\cos^2 x + \sin x + 1 = 0$ lies between:

$(3\pi/4, 5\pi/4)$

$(5\pi/4, 7\pi/4)$

$(\pi/4, 3\pi/4)$

$(-\pi/4, \pi/4)$

$(1 + \cot A + \tan A) (\sin A - \cos A) \frac{\sin A \cos A}{\sin^3 A - \cos^3 A}$ का मान क्या है-

- 0
- 1
- 1
- 21

यदि $\sin x + \sin^2 x = 1$ है, तो $\cos^8 x + 2 \cos^6 x + \cos^4 x$ का मान क्या है?

- 0
- 1
- 2
- 4

If $\sin m + \sin n = p$, $\cos m + \cos n = q$, then find the value of $\sin m \times \sin n + \cos m \times \cos n$.

(a) $p^2 + q^2 - 2$

(b) $(p^2 + q^2 - 2)/2$

(c) $p + q - pq$

(d) $p + q + pq$

$\sin(x-y) = 1/2$ and $\cos(x+y) = 1/2$, तो $\sin x \cos x + 2\sin^2 x + \cos^3 x \sec x$ का मान क्या है?

1

2

$\frac{3}{4}$

0

$$\frac{\sin 8\theta \cos \theta - \sin 6\theta \cos 3\theta}{\cos 2\theta \cos \theta - \sin 3\theta \sin 4\theta}$$

(a) $\cot \theta$

(b) $\tan \theta$

(c) $\tan 2\theta$

(d) $\cot 2\theta$

यदि $\left\{ \left(\frac{\sec \theta - 1}{\sec \theta + 1} \right) \right\}^n = \operatorname{cosec} \theta - \cot \theta$ है, तो $n = ?$

(a) 1

(b) 0.5

(c) -1

(d) -0.5

$(2\cos^2\theta - 1) \left[\frac{1+\tan\theta}{1-\tan\theta} + \frac{1-\tan\theta}{1+\tan\theta} \right]$ का मान क्या है?

- (a) 2
- (b) 0
- (c) $\sqrt{3}$
- (d) 1

If $\cos A + \cos B + \cos C = 3$, then what is the value of $\sin A + \sin B + \sin C$?

- (a) 1
- (b) 2
- (c) 0
- (d) -1

If $\cos A = \sin^2 A$, and

$a \sin^{12} A + b \sin^{10} A + c \sin^8 A + \sin^6 A = 1$, then $a + b + c = ?$

(a) 7

(b) 8

(c) 9

(d) 6

Thankyou

Theory

RADIANS AND DEGREES

DEGREES TO RADIANS

- Since $180^\circ = \pi$ radians
- Therefore, $1^\circ = \frac{\pi}{180}$ radians
- Hence, $30^\circ = \frac{\pi}{180} \times 30 = \frac{\pi}{6}$ radians

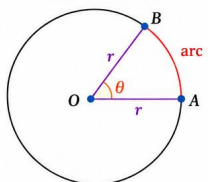
$45^\circ = \frac{\pi}{180} \times 45 = \frac{\pi}{4}$ radians

$60^\circ = \frac{\pi}{180} \times 60 = \frac{\pi}{3}$ radians

$90^\circ = \frac{\pi}{180} \times 90 = \frac{\pi}{2}$ radians

RULE

The number of radians in an angle subtended by an arc of a circle at the centre is equal to $\frac{\text{arc}}{\text{radius}}$



or, $\theta = \left(\frac{l}{r}\right)^c$ (l is arc length and r is radius)

COMMON CONVERSIONS

Degree	30°	45°	60°	90°	120°	135°	150°	180°	270°	360°
Radian	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π	$\frac{3\pi}{2}$	2π

HOW MANY DEGREES IN ONE RADIAN?

- We have,
 π radians = 180°
 $\therefore 1 \text{ radian} = \frac{180^\circ}{\pi} = \left(\frac{180}{22} \times 7\right)^\circ$
 $= 57^\circ 16' 22''$ (approx.)

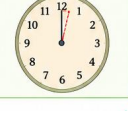
HOW MANY RADIANS IN ONE DEGREE?

- We have,
 $180^\circ = \pi$ radians
 $\therefore 1^\circ = \frac{\pi}{180}$ radians
 $= \left(\frac{22}{7 \times 180}\right)$ radian = 0.01746 radian.

SOME USEFUL POINTS

- The angle between two consecutive digits in a clock is
 $30^\circ = \frac{\pi}{6}$ radians.
- The hour hand rotates through an angle of 30° in one hour i.e.
 $\left(\frac{1}{2}\right)^\circ$ in one minute.
- The minute hand rotates through an angle of 6° in one minute.
 $6^\circ = \frac{\pi}{30}$ radians.





Q.1

The degree measure of 1 radian (taking $\pi = 22/7$) is

1 रेडियन का डिग्री माप क्या होगा, यदि $\pi = 22/7$ लिया जाए?

- (a) $57^{\circ}61'22''$ (approx.)
- (b) $57^{\circ}16'22''$ (approx.)
- (c) $57^{\circ}22'16''$ (approx.)
- (d) $57^{\circ}32'16''$ (approx.)

Q.2

Angle 54° is equivalent to (in radians) ?

54° कोण रेडियन में किसके बराबर होगा?

- (a) $9\pi/10$
- (b) $7\pi/10$
- (c) $\pi/10$
- (d) $3\pi/10$

Q.3

In circular measure, the value of the angle $11^{\circ}15'$ is ?

वृत्तीय माप यानी रेडियन में $11^{\circ}15'$ कोण का मान क्या होगा?

- (a) $\pi/16$
- (b) $\pi/8$
- (c) $\pi/4$
- (d) $\pi/12$

Q.4

If $0 \leq \theta \leq \pi/2$ and $\sec^2\theta + \tan^2\theta = 7$, then θ is ?

यदि $0 \leq \theta \leq \pi/2$ और $\sec^2\theta + \tan^2\theta = 7$ है, तो θ का मान क्या होगा?

- (a) $5\pi/12$ radian
- (b) $\pi/3$ radian
- (c) $\pi/5$ radian
- (d) $\pi/6$ radian

Q.5

The sum of two angles is 155° and their difference is $\pi/2$.
The value of the greater angle (in radians) is ?

दो कोणों का योग 155° है और उनका अंतर $\pi/2$ है। बड़े कोण का मान रेडियन में क्या होगा?

- (a) $53\pi/72$
- (b) $47\pi/72$
- (c) $49\pi/72$
- (d) $51\pi/72$

Q.6

By decreasing 17° from each angle of a triangle, the ratio of their angles is 3:4:7. The radian measure of the greatest angle is ?

किसी त्रिभुज के प्रत्येक कोण में से 17° घटाने पर उनके कोणों का अनुपात 3:4:7 हो जाता है। सबसे बड़े कोण का रेडियन माप क्या होगा?

- (a) $153\pi/360$
- (b) $167\pi/180$
- (c) $163\pi/180$
- (d) $163\pi/360$

Q.7

In a triangle ABC, Angle ABC = 75° and Angle ACB = $\pi/4$.
The circular measure of Angle BAC is ?

त्रिभुज ABC में कोण ABC = 75° और कोण ACB = $\pi/4$ है।
कोण BAC का वृत्तीय माप यानी रेडियन माप क्या होगा?

- (a) $5\pi/12$ radian
- (b) $\pi/3$ radian
- (c) $\pi/6$ radian
- (d) $\pi/2$ radian

Q.8

The area of the sector of a circle of radius 28 cm is 112 cm^2 . Find the length of the corresponding arc of the sector ?

28 सेमी त्रिज्या वाले एक वृत्त के त्रिज्यखंड का क्षेत्रफल 112 सेमी^2 है। उस त्रिज्यखंड की संबंधित चाप की लंबाई ज्ञात कीजिए।

- (a) 4 cm
- (b) 8 cm
- (c) 6 cm
- (d) 5 cm

Q.9

A sector of a circle has a central angle of 120° and a radius of 7 cm. Another sector of the same circle has a central angle of $2\pi/3$ radians. What is the ratio of the area of the first sector to the area of the second sector ?

एक वृत्त के एक त्रिज्यखंड का केंद्रीय कोण 120° और त्रिज्या 7 सेमी है। उसी वृत्त के दूसरे त्रिज्यखंड का केंद्रीय कोण $2\pi/3$ रेडियन है। पहले त्रिज्यखंड के क्षेत्रफल और दूसरे त्रिज्यखंड के क्षेत्रफल का अनुपात क्या होगा?

- (a) 3:5
- (b) 2:3
- (c) 1:1
- (d) 4:5

Q.10

An arc of length 23.1 cm subtends an 18° angle at the centre. What is the area of the circle ? [Use $\pi = 22/7$]

23.1 सेमी लंबाई की एक चाप केंद्र पर 18° का कोण बनाती है। वृत्त का क्षेत्रफल क्या होगा? [$\pi = 22/7$ का प्रयोग कीजिए]

- (a) 16978.50 cm^2
- (b) 16988.50 cm^2
- (c) 16878.50 cm^2
- (d) 16798.50 cm^2